

Xiwen Huang

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Education

Southern University of Science and Technology (SUSTech)

Shenzhen, China

Bachelor of Engineering in Information Engineering

Expected Jun. 2027

- GPA: 3.76/4.0; Honors Class
- Selected coursework: Analog Circuits Laboratory (100/100), Digital Circuits Laboratory (99/100), Fundamentals of Electric Circuits (95/100), Digital Signal Processing (95/100), Digital Circuits (92/100), Signals and Systems (92/100), Engineering Electromagnetics (90/100), Data Communications and Networking (90/100), Introduction to Computer Programming (96/100), Calculus II (94/100)

University of Pennsylvania

Philadelphia, USA

Undergraduate Exchange Student

Jan. 2026 - May 2026

- GPA: 4.0/4.0
- Selected coursework: Machine Learning (A), Foundations of AI: Deep Learning with Applications (A), Musical Interfaces and Robotics (A+)

Research Interests

My research interests lie in reinforcement learning for robotics, with an emphasis on sim-to-real transfer, controller-aware policy deployment, and reliable behavior in contact-rich tasks.

Honors & Awards

2026, 2024: SUSTech Study Abroad Scholarship

2025: Honorable Mention, Mathematical Contest in Modeling (MCM) — Team Leader

2024: SUSTech, Department of Electronics and Electrical Engineering - Wo Long Elite Class (Top 8%, Honors Class)

2024: Sole Winning Team, NVIDIA Program of Blended Learning, MIT xPRO

2024: SUSTech Outstanding Student

2024: SUSTech Excellence Scholarship

Publications

CA-LRR: Controller-Aware Low-Rank Residual Refinement for Contact-Rich Aerial Manipulation

Second author; submitted to the IEEE International Conference on Robotics and Automation (ICRA 2027).

Research & Project Experience

CA-LRR: Sim-to-Real Transfer for Contact-Rich Aerial Manipulation

Research Assistant; Westlake University; Advisor: Shiyu Zhao

Jun. 2026 - Sep. 2026

- Developed and trained reinforcement learning policies in Isaac Lab for coordinated aerial base-arm control with an 11-dimensional action space, tuning network architectures, training hyperparameters, and reward functions, and configuring simulation friction and randomized initial states.
- Engineered ROS-based deployment with Vicon pose feedback and PreGME flight control, decoupling 8 Hz policy inference from 50 Hz setpoint generation across separate ROS nodes. Applied exponential moving-average filtering to base actions, time-based linear interpolation of base position and yaw, and velocity- and acceleration-limited joint interpolation to smooth command execution, while addressing timing and dynamics discrepancies between simulation and hardware.
- Contributed to low-rank residual refinement around a frozen reference policy using a five-dimensional SVD basis from successful simulation rollouts; implemented tracking-error-based residual scaling and direction-preserving enforcement of action and joint limits. The resulting method increased mean task success from 44.91% to 56.72% across nine simulated load-slope conditions. Hardware experiments demonstrated pushing a cart with a total mass of 35.54 kg and, in a separate test, pushing across a 2 cm-high uneven surface.

LLM Hallucination Research

Research Assistant; University of Pennsylvania; Advisor: Enric Boix

Feb. 2026 - May 2026

- Fine-tuned TinyLlama-1.1B on SQuAD 2.0 using PEFT and a custom abstention objective to encourage the model to withhold answers when uncertain.
- Reduced incorrect responses without abstention by 12% relative to the baseline on a held-out test split.

Universal Vessel-like Structure Segmentation

Research Assistant; SUSTech; Advisor: Xiaoying Tang Jul. 2025 - Dec. 2025

- Developed an nnU-Net-based model with prompt fusion and dual attention to segment 10+ vessel-like structures across multiple imaging modalities in a single network.
- Improved mean Dice from 84.92 to 85.79 (+0.87) and mean cIDice from 89.58 to 91.38 (+1.80) across seven 2D datasets, outperforming single-task nnU-Net baselines.

Quadruped Locomotion with Analytical Policy Gradients

Research Assistant; SUSTech; Advisor: Jiankun Wang Feb. 2025 - Jun. 2025

- Implemented analytical policy gradient (APG) training in MuJoCo XLA (MJX) for quadruped locomotion and residual motion policies, demonstrating higher sample efficiency than PPO in simulation.
- Adapted the ANYmal simulation environment to the Unitree AI platform and built interactive visualization and parameter persistence tools to support reproducible testing and deployment.

Professional Experience

Shennan Circuits Co., Ltd Shenzhen, China

Intern, IT Process and Control Department Jan. 2025 - Feb. 2025

- Led development of a full-stack shuttle-bus information system with an HTML/CSS frontend and a Java backend, providing real-time arrival information to 50+ employees.
- Designed and managed an SQL database for bus routes, schedules, and real-time passenger waiting data.

Technical Skills

Programming: Python, C, C++, Java, MATLAB, HTML/CSS, SQL, LaTeX, LabVIEW

Frameworks & Tools: PyTorch, PEFT, MuJoCo, ROS, PX4, Gazebo

Hardware & Embedded: Raspberry Pi, Embedded Systems, Circuit Design

Languages

Mandarin Chinese: Native

English: TOEFL iBT 5.5

Japanese: Elementary

Spanish: Elementary